Studio 4

Abstract Classes

Objectives

To introduce the concept of abstract classes and how they can be used with inheritance.

Exercise 1

1. How is an object actually stored?

2. What is an abstract class?

3. What is an abstract method?

4. You have decided to create your own side scrolling shooter game and want to start with some basic game mechanics of moving a player and creating and moving enemies.

   Look at the class diagram for Ex 1 on Moodle and implement the Sprite class.

5. Implement the PlayerShip class and set the SHIP_WIDTH constant to 50 and the SHIP_HEIGHT constant to 10.

   The drawSprite method should draw a rectangle for the ship centred around it's x and y position in the correct colour.

   The moveSprite method should move the y position of the ship by the amount passed to the method.

6. Implement the EnemyShip class and set the SIZE constant to 40.

   The drawSprite method should draw an ellipse for the enemy ship centred around it's x and y position in the correct colour.

   The moveSprite method should move the x position ~~y~~ position of the enemy ship to the left by the amount passed to the method.

7. In the setup method of the sketch code set the size of the sketch window and create a new player ship at x position 50 and y position half way down the sketch window.

8. In the draw method of the sketch window set the background colour of the sketch window to black then draw the player ship and test that you can see the player ship.

9. Create the keyPressed method and if the up arrow key is pressed move the player ship by -4 pixels and if the down arrow key is pressed move it by 4 pixels. Test that you can move your player ship.

10. Create an arrayList to store enemy ship objects at class scope level. In the draw method write the code that will draw each enemy ship in the list.

11. Then write the code so that on every 30th frame a new enemy ship is added to the array list at the right hand edge of the sketch window and a random y location. Test that enemy ships are being created.

12. Then write the code that will move all of the enemy ships by 4 pixels. Test that everything is moving correctly now.

Exercise 2

1. You have decided to create your own action RPG game like the stupendously awesome Grim Dawn and make your fortune. You will need to create the basic classes needed to store info about the various items in the game. There are initially two classes of items, Weapon and Armour and for each class there are different types.

   For instance, there are swords, maces and ranged weapons. There are chest armour, leg armour and head armour.

   Create a new sketch and create the classes based on the class diagram posted on Moodle.

Note: In the toString method of the weapon class format the attack speed value to have 1 digit to the left of the decimal point and 2 digits to the right of the decimal point using the **nf()** method.

2. Download the Items.csv file from Moodle and add it your sketch. Create a list of items at class scope level. Write the code in the setup method to process the csv file and if the length of the split array is 5 then add a new Weapon object to the list, if the length of the split array is 4 then add a new Armour object to the list otherwise display an error message.

3. After the file has been processed display each object's information to the console window.

Exercise 3 (Studio Handin Exercise)

1. Create a new sketch and you are going to create classes to allow for different types of buttons. Look at the class diagram for this exercise on Moodle.

2. Implement the Button class.

3. Implement the RectangleButton class.

   In the constructor set the font to a font of your choosing and the size to 14.

   In the drawButton method use the textAlign method to make the caption centred in the button and the text colour should be white.

   The isClicked method should be the same as in the previous studio exercise.

4. Implement the CircleButton class.

   When implementing the isClicked method the condition for the if statement is:

   ```
   (x - _centre.x)^2 + (y - _centre.y)^2 < (size/2)^2
   ```

   If the condition above is true then return true else return false.

5. Create a reference variable for each button at class scope level in the sketch code and then create new objects for them in the setup method. Then in the draw method draw the two buttons and check that they appear in the sketch window.

6. Create the mousePressed method that checks each button if it is clicked on and if it is then displays a message in the console window.

7. Download the buttons.csv file from Moodle and add it to your sketch. The csv file has the following format if the line is for a rectangle button:

```
x,y,caption,red,green,blue,width,height
```

It has the following line for a circle button:

```
x,y,red,green,blue,size
```

Write the code to create the appropriate button objects using the data in the csv file and store in an array list. You will need to use the red, green and blue values to create a colour value using the color method. In the draw method in the sketch code draw all the buttons to the sketch window.

8. In the mousePressed method write the code that will check all of the buttons and write a message to the console window if a button has been clicked on.